

## SQL Exercise 2: Aggregate Functions and Grouping.

### 1) COUNT() Function

Write a SQL query to find the total number of employees in the company.

```
SELECT COUNT(ID) AS number of employees  
FROM employees;
```

| Number of employees |
|---------------------|
| 10                  |

### 2) SUM() Function

Write a SQL query to find the total salary paid to all employees in the IT department

```
SELECT department,  
SUM(salary) AS Total_salary  
FROM employees,  
WHERE department = 'IT';
```

| Department | Total - Salary |
|------------|----------------|
| IT         | 220 000        |

### 3) AVG Function

Write a SQL query to calculate the average salary of employees in the HR department.

```
SELECT department  
AVG(salary) AS average_salary  
FROM employees  
WHERE department = 'HR';
```

| Department | Average-salary |
|------------|----------------|
| HR         | 49500          |

➤ MIN() and MAX() functions

Write a SQL query to find the highest and lowest salary in the company.

```
SELECT MIN(salary) AS lowest_salary
FROM employees;
```

| Lowest_salary |
|---------------|
| 48000         |

```
SELECT MAX(salary) AS highest_salary
FROM employees;
```

| Highest_salary |
|----------------|
| 62000          |

5) GROUP BY statement.  
Write a SQL query to group employees by department and display the total salary paid in each department.

```
SELECT department,
SUM(salary) AS total_salary
FROM employees
```

```
GROUP BY department;
```

| Department | Total-salary |
|------------|--------------|
| IT         | 220 000      |
| Finance    | 119 000      |
| HR         | 99 000       |
| Marketing  | 105 000      |

## 6) GROUP BY and COUNT()

Write a SQL query to count how many employees work in each city

```
SELECT city,  
        COUNT(ID) AS total number of employees  
FROM employees  
GROUP BY city;
```

| Total number of employees | City          |
|---------------------------|---------------|
| 2                         | Los Angeles   |
| 2                         | New York      |
| 3                         | Chicago       |
| 1                         | Houston       |
| 2                         | San Francisco |

7) GROUP BY and ORDER BY  
Write a SQL query to group employees by department, calculate the average salary in each department, and order the results in descending order of average salary.

```
SELECT department,  
        AVG(salary) AS average_salary  
FROM employees  
GROUP BY department  
ORDER BY average_salary DESC;
```

### 8) HAVING Clause.

Write a SQL query to find departments where the total salary paid exceeds 100000.

(Use GROUP BY and HAVING)

```
SELECT department;  
       SUM(salary) AS total_salary  
FROM employees  
GROUP BY department  
HAVING SUM(salary) > 100000;
```

### 9) Combining GROUP BY, HAVING and ORDER BY

Write a SQL query to list cities where more than one employee works, ordered by the number of employees in descending order.

```
SELECT city,  
       COUNT(*) AS total_employee  
FROM employees  
GROUP BY city  
HAVING COUNT(*) > 1  
ORDER BY total_employee DESC;
```

## 10) Combining Aggregate Functions

Write a SQL query to find the department with the highest average salary.

```
SELECT department,  
       AVG (salary) AS average_salary  
FROM   employees  
GROUP BY department  
ORDER BY average_salary DESC  
LIMIT 1 ;
```

| Department | Highest salary | Average - salary |
|------------|----------------|------------------|
| IT         | 220000         | 55000            |